

CS 156

Intro to AI



Inference in Bayes Nets

Probabilistic Reasoning Over Time

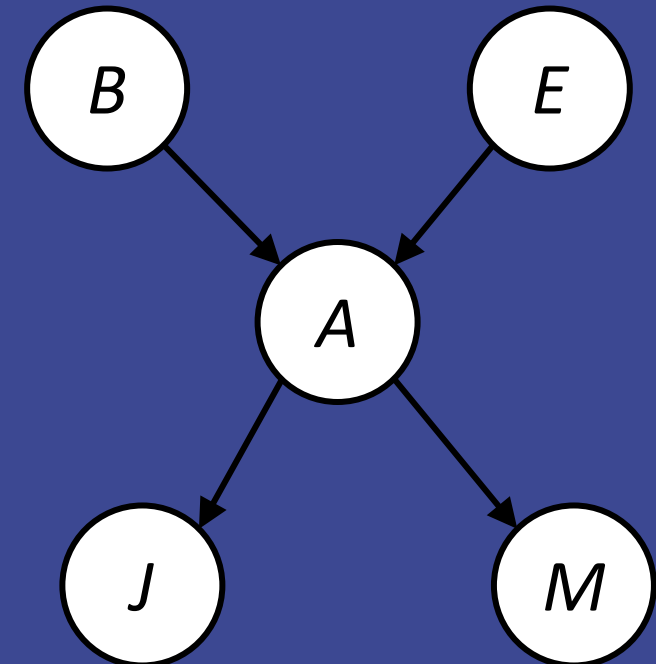
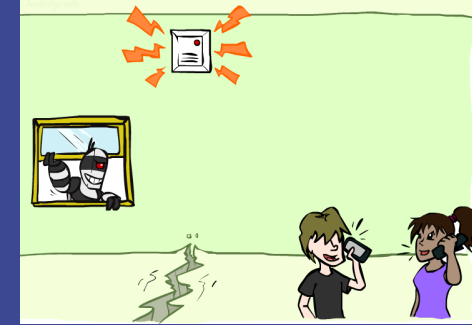
These slides are based on the slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.
The artwork is by Ketrina Yim

Today

- ▶ Inference by Enumeration in Bayes Nets
- ▶ Homework 6
- ▶ Markov Chains
- ▶ Hidden Markov Models

Inference by Enumeration in Bayes' Net

- ▶ John and Mary called.
- ▶ Was there a burglary?
- ▶ Query Variable?
 - B
- ▶ Evidence Variables?
 - J, M
- ▶ Hidden Variables?
 - E, A



We need to compute: $P(+b \mid +j, +m)$

Inference by Enumeration in Bayes' Net

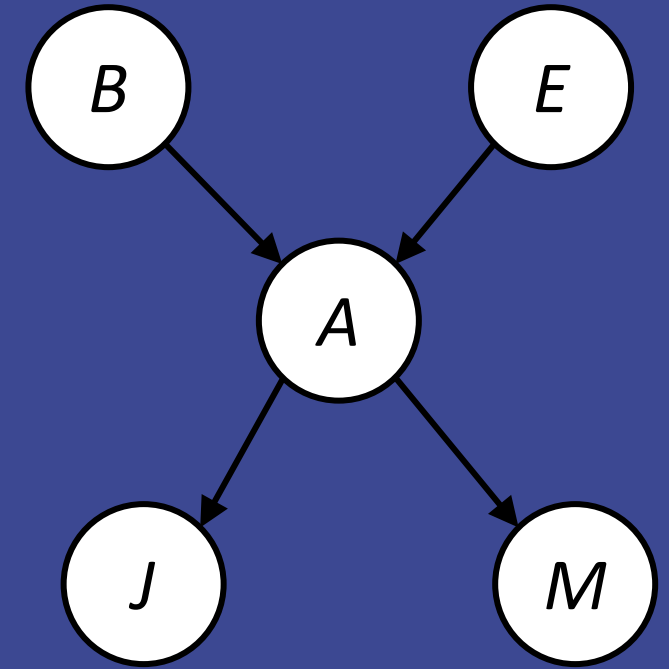
By definition of conditional probability:

$$P(+b \mid +j, +m) = \frac{P(+b, +j, +m)}{P(+j, +m)}$$

We also know that $P(X) = P(X, +y) + P(X, -y)$:

$$P(+b, +j, +m) = ?$$

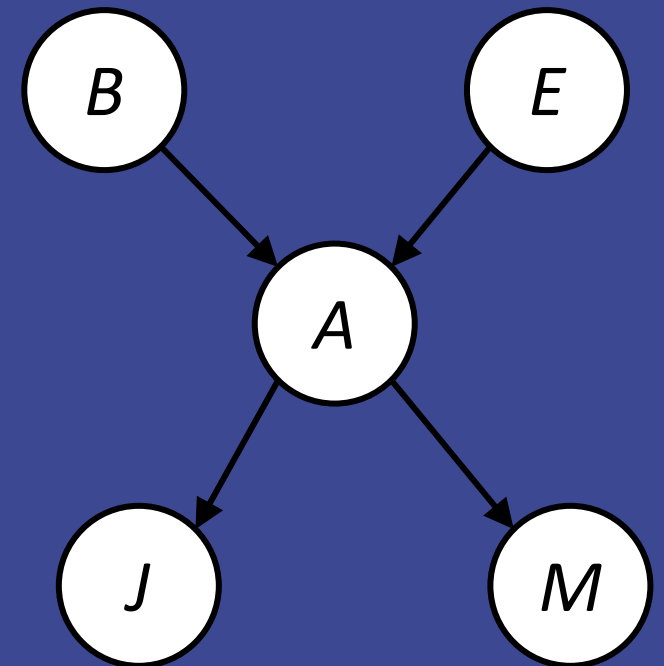
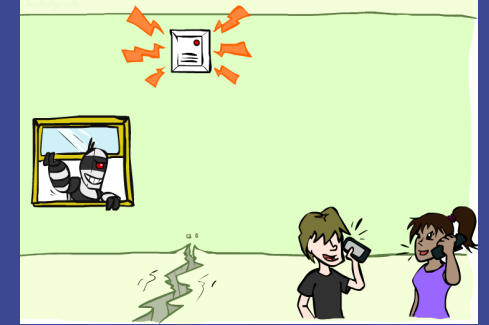
$$\begin{aligned} P(+b, +j, +m) &= P(+b, +j, +m, +e) + P(+b, +j, +m, -e) \\ &= P(+b, +j, +m, +e, +a) + P(+b, +j, +m, +e, -a) + \\ &\quad P(+b, +j, +m, -e, +a) + P(+b, +j, +m, -e, -a) \end{aligned}$$



Example: Alarm Network

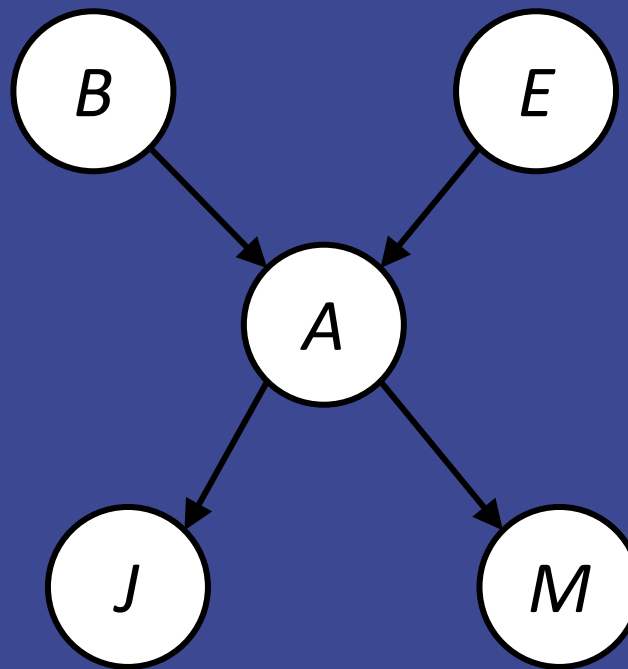
What is $P(+b, +j, +m, +e, +a)$?

- A. $P(+b) P(+j) P(+m) P(+e) P(+a)$
- B. $P(+b) P(+e) P(+a | +b, +e) P(+j | +a) P(+m | +a)$
- C. $P(+j) P(+m) P(+a | +j, +m) P(+b | +a) P(+e | +a)$
- D. $P(+b) P(+e) P(+a | +b) P(+a | +e) P(+j | +a) P(+m | +a)$
- E. $P(+b) P(+j) P(+m) P(+e) P(+a | +b, +e, +j, +m)$



Example: Alarm Network

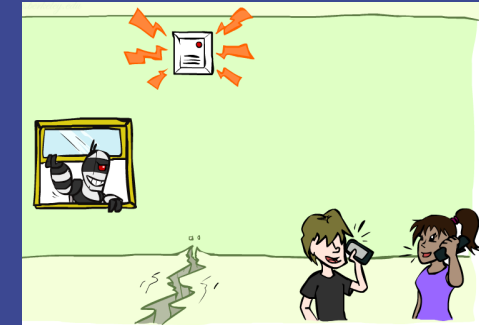
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



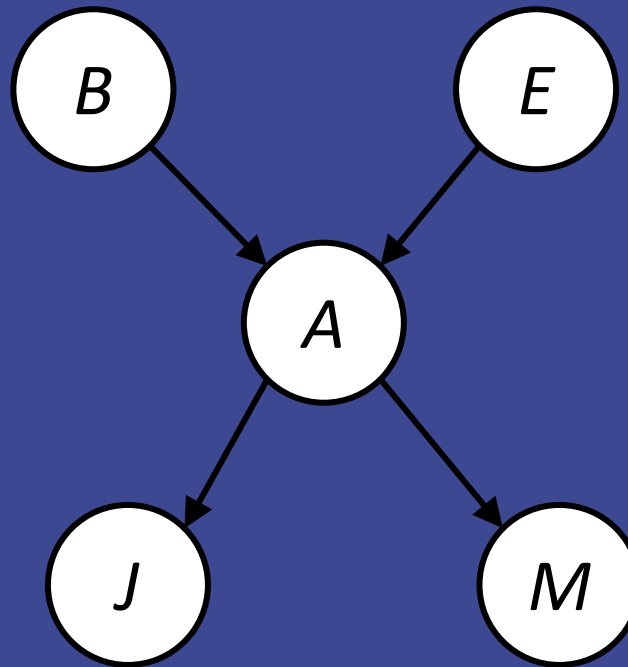
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, +j, +m, +e, +a) =$$

$$P(+b) P(+e) P(+a | +b, +e) P(+j | +a) P(+m | +a) =$$

Example: Alarm Network

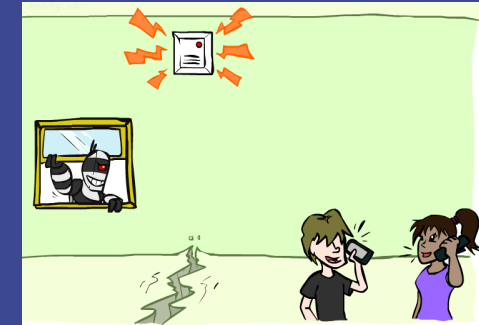
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95



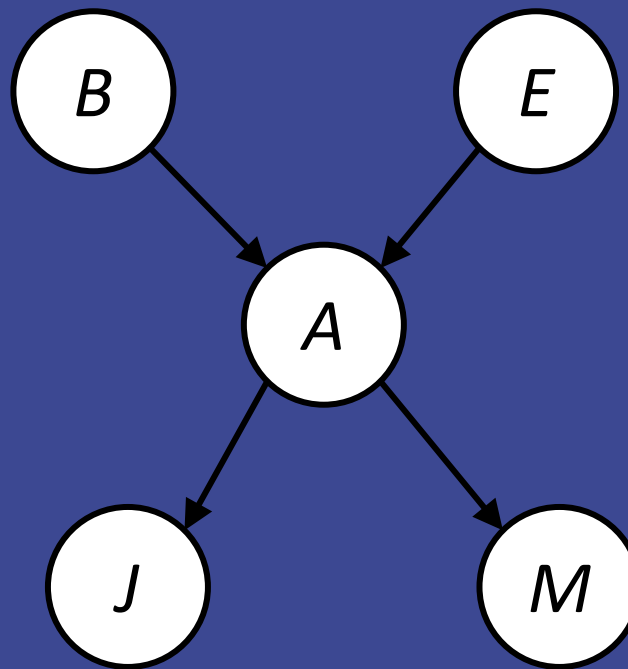
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$P(+b, +j, +m, +e, +a) =$

$P(+b) P(+e) P(+a|+b, +e) P(+j|+a) P(+m|+a) =$
0.001

Example: Alarm Network

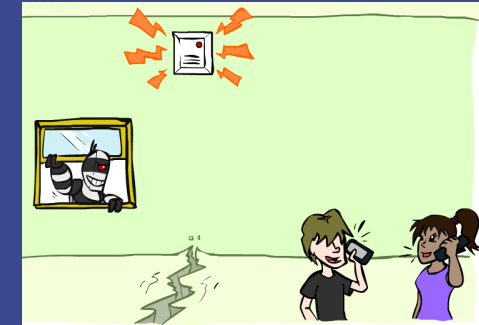
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

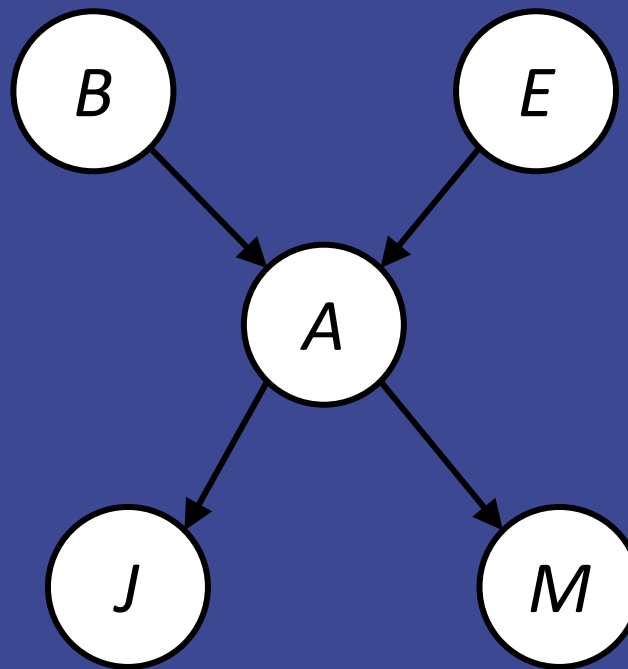


B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 &P(+b, +j, +m, +e, +a) = \\
 &P(+b) P(+e) P(+a | +b, +e) P(+j | +a) P(+m | +a) = \\
 &0.001 \times 0.002
 \end{aligned}$$

Example: Alarm Network

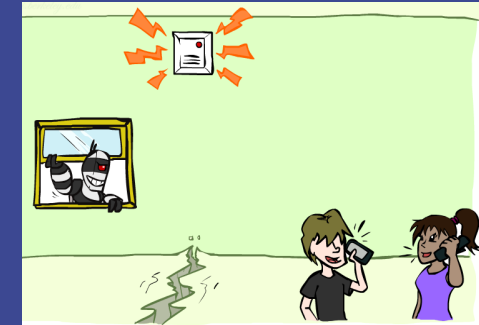
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95



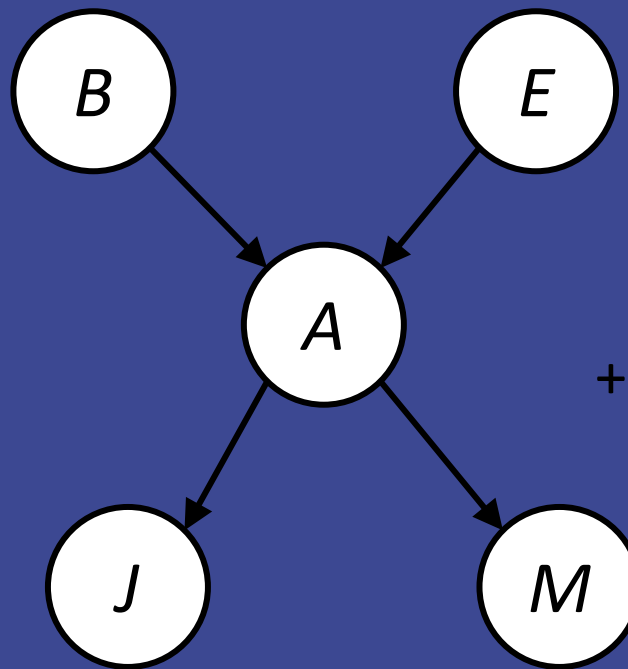
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, +j, +m, +e, +a) = P(+b) P(+e) P(+a | +b, +e) \\ 0.001 \times 0.002$$

iClicker:
What is $P(+a | +b, +e)$?

Example: Alarm Network

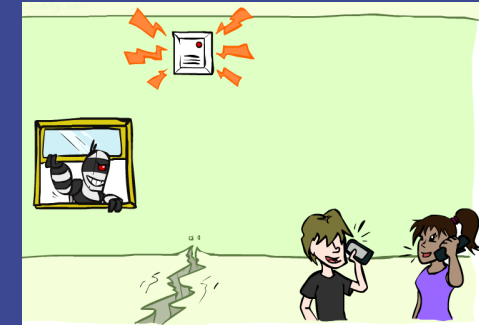
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

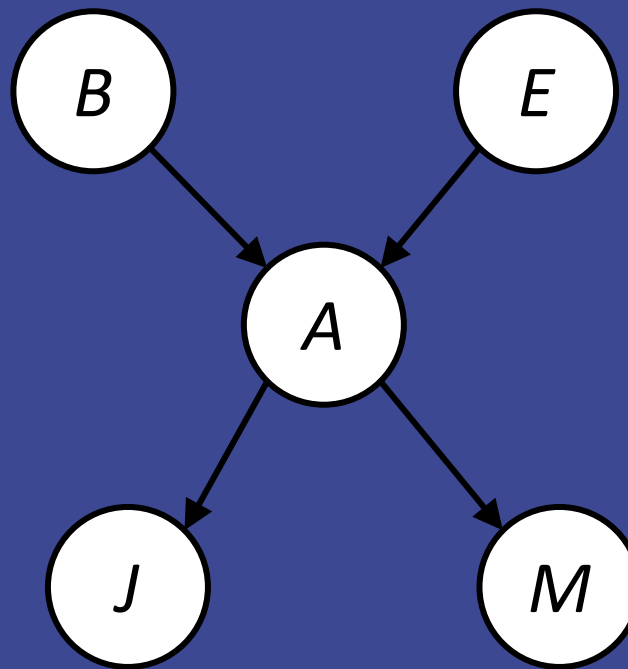


B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 &P(+b, +j, +m, +e, +a) = \\
 &P(+b) P(+e) P(+a|+b, +e) P(+j|+a) P(+m|+a) = \\
 &0.001 \times 0.002 \times 0.95
 \end{aligned}$$

Example: Alarm Network

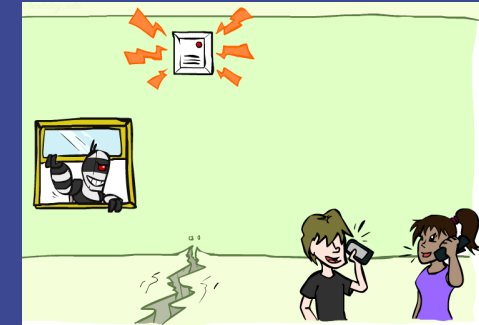
B	P(B)
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-b	0.999



E	P(E)
+e	0.002
-e	0.998

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-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

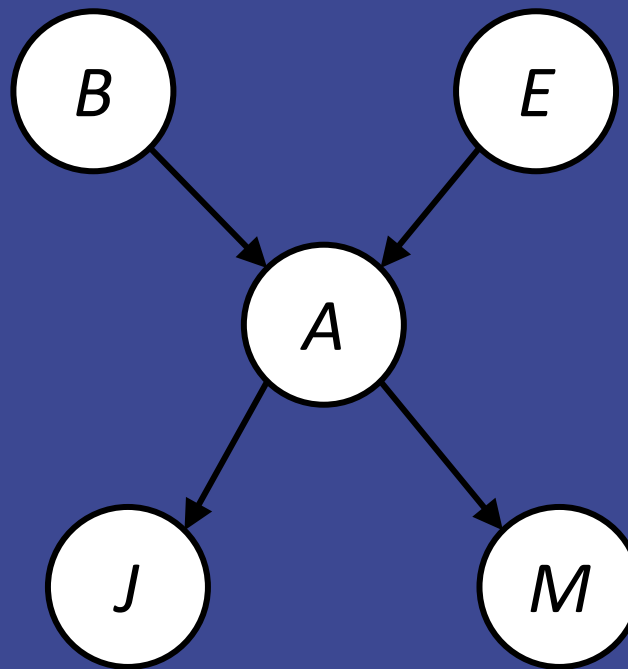
$$P(+b, +j, +m, +e, +a) =$$

$$P(+b) P(+e) P(+a|+b, +e) P(+j|+a) P(+m|+a) =$$

$$0.001 \times 0.002 \times 0.95 \times 0.9 \times 0.7 = 1.2 \times 10^{-6}$$

Example: Alarm Network

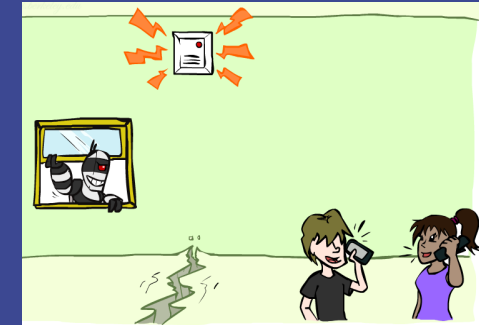
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95



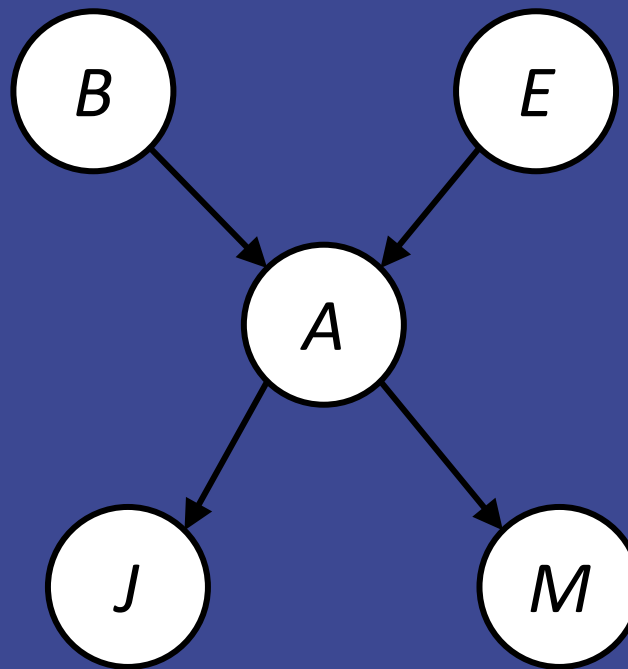
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, +j, +m, +e, -a) =$$

$$P(+b) P(+e) P(-a | +b, +e) P(+j | -a) P(+m | -a) =$$

Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

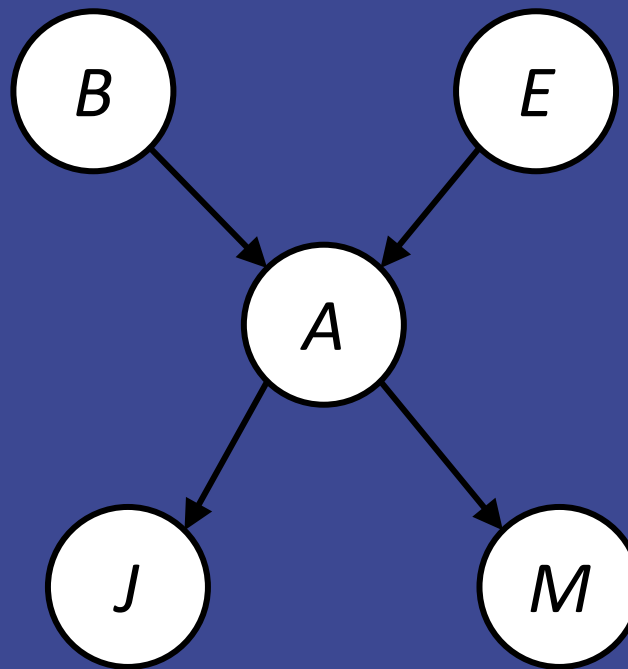
$$P(+b, +j, +m, +e, -a) = P(+b) P(+e) P(-a | +b, +e) P(+j | -a)$$

$$0.001 \times 0.002 \times 0.05$$

iClicker:
What is $P(+j | -a)$?

Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, +j, +m, +e, -a) =$$

$$P(+b) P(+e) P(-a | +b, +e) P(+j | -a) P(+m | -a) =$$

$$0.001 \times 0.002 \times 0.05 \times 0.05 \times 0.01 = 5 \times 10^{-11}$$

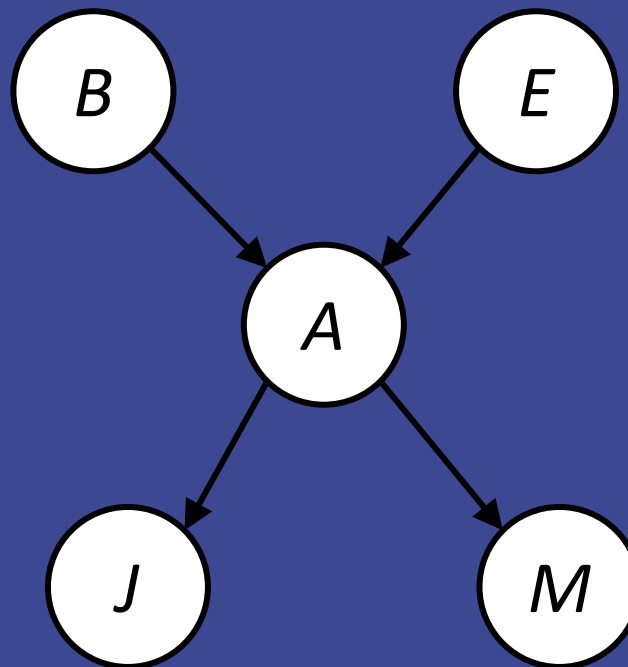
Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

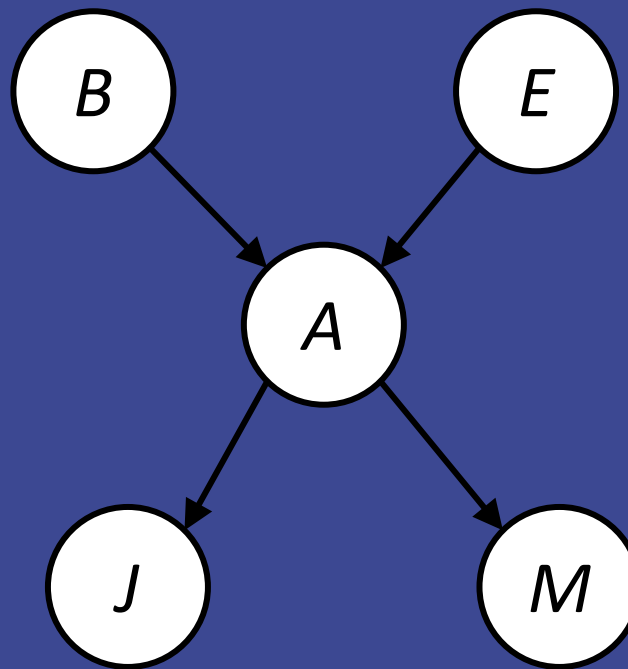
$$P(+b, +j, +m, -e, +a) =$$

$$P(+b) P(-e) P(+a | +b, -e) P(+j | +a) P(+m | +a) =$$

$$0.001 \times 0.998 \times 0.94 \times 0.9 \times 0.7 = 0.000591$$

Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

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+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

B	E	A	P(A B,E)
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+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, +j, +m, -e, -a) =$$

$$P(+b) P(-e) P(-a | +b, -e) P(+j | -a) P(+m | -a) =$$

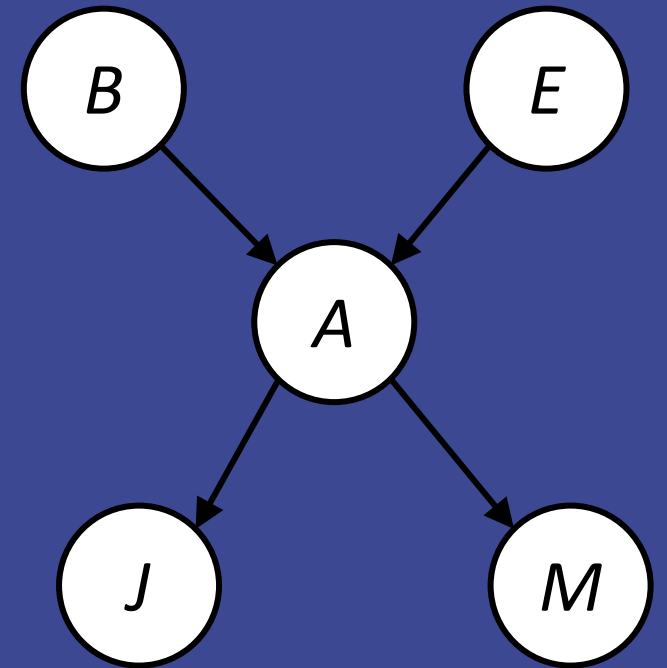
$$0.001 \times 0.998 \times 0.06 \times 0.05 \times 0.01 = 3 \times 10^{-8}$$

Inference by Enumeration in Bayes' Net

Putting it all together:

$$P(+b \mid +j, +m) = \frac{P(+b, +j, +m)}{P(+j, +m)}$$

$$\begin{aligned} P(+b, +j, +m) &= P(+b, +j, +m, +e, +a) + \\ &\quad P(+b, +j, +m, +e, -a) + \\ &\quad P(+b, +j, +m, -e, +a) + \\ &\quad P(+b, +j, +m, -e, -a) \\ &= 0.000592 \end{aligned}$$

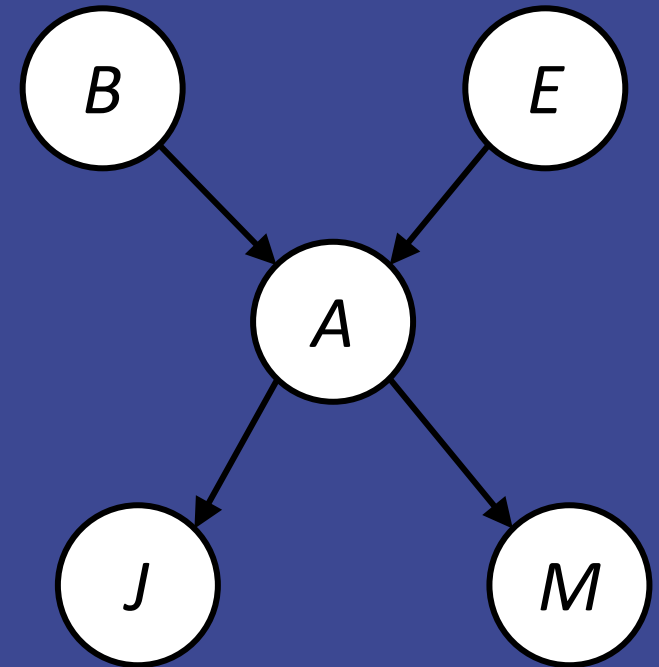


Inference by Enumeration in Bayes' Net

Similarly, we can calculate: $P(-b \mid +j, +m)$

$$P(-b \mid +j, +m) = \frac{P(-b, +j, +m)}{P(+j, +m)}$$

$$\begin{aligned} P(-b, +j, +m) &= P(-b, +j, +m, +e, +a) + \\ &\quad P(-b, +j, +m, +e, -a) + \\ &\quad P(-b, +j, +m, -e, +a) + \\ &\quad P(-b, +j, +m, -e, -a) \\ &= 0.001492 \end{aligned}$$



Inference by Enumeration in Bayes' Net

$P(-b, +j, +m)$ and $P(+b, +j, +m)$ are the joint probabilities.

We still need to compute the conditional probabilities.

$$P(-b \mid +j, +m) = \frac{P(-b, +j, +m)}{P(+j, +m)} = \frac{0.001492}{P(+j, +m)}$$

$$P(+b \mid +j, +m) = \frac{P(+b, +j, +m)}{P(+j, +m)} = \frac{0.000592}{P(+j, +m)}$$

We normalize:

$$\begin{aligned} Z &= P(-b, +j, +m) + P(+b, +j, +m) \\ &= 0.001492 + 0.000592 = 0.002084 \end{aligned}$$

$$P(-b \mid +j, +m) = 0.716$$

$$P(+b \mid +j, +m) = 0.284$$

Bayes Nets Summary

- ▶ Bayes nets **compactly** encode joint distributions
- ▶ **Guaranteed independences** of distributions can be deduced from BN graph structure
- ▶ **Inference by enumeration** is the basic **exact inference** method for Bayes nets. There are techniques (variable elimination) to make it more efficient
- ▶ There are also some approximate inference methods that include randomized sampling

Homework 6: beliefs.py

```
class Belief(object):
```

```
    """
```

Belief class used to track the belief distribution based on the sensing evidence we have so far.

Arguments:

size (int): the number of rows/columns in the grid

Attributes:

open (set of tuples): set containing all the positions that have not been observed so far.

current_distribution (dictionary): probability distribution based on the evidence observed so far.

The keys of the dictionary are the possible grid positions

The values represent the (conditional) probability that the treasure is found at that position given the evidence (sensor data) observed so far.

```
    """
```

Homework 6: beliefs.py

```
class Belief(object):
    def __init__(self, size):
        # Initially all positions are open - have not been observed
        self.open = {(x, y) for x in range(size) for y in range(size)}

        {(4, 0), (3, 4), (4, 3), (3, 1), (0, 2), (2, 2), (1, 0), (1, 3), (4, 2),
        (3, 0), (3, 3), (0, 1), (2, 4), (1, 2), (0, 4), (2, 1), (3, 2), (4, 1),
        (4, 4), (0, 0), (1, 1), (0, 3), (2, 0), (1, 4), (2, 3)}

        # Initialize to a uniform distribution
        self.current_distribution = {pos: 1 / (size ** 2) for pos in self.open}

        {(4, 0): 0.04, (3, 4): 0.04, (4, 3): 0.04, (3, 1): 0.04, (0, 2): 0.04, (2,
        2): 0.04, (1, 0): 0.04, (1, 3): 0.04, (4, 2): 0.04, (3, 0): 0.04, (3, 3):
        0.04, (0, 1): 0.04, (2, 4): 0.04, (1, 2): 0.04, (0, 4): 0.04, (2, 1):
        0.04, (3, 2): 0.04, (4, 1): 0.04, (4, 4): 0.04, (0, 0): 0.04, (1, 1):
        0.04, (0, 3): 0.04, (2, 0): 0.04, (1, 4): 0.04, (2, 3): 0.04}
```

	SONAR		DIVE	
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04

Inference in Homework 6: update

Prior distribution over treasure location: $P(T)$

- Initially uniform
- ▶ Sonar reading model: $P(C|T)$
 - Given: we know what our sensors do (use `model.pcolorgivendist`)
- ▶ C = sonar color measured at (1, 3)
- ▶ $P(C = \text{green} | T)$ from model
- ▶ We can calculate the posterior distribution $P(T|C)$ over treasure locations given a sonar reading using Bayes' rule:

$$P(T|C) = \frac{P(T) P(C|T)}{P(C)} \propto P(T) P(C|T)$$

$\propto$: is proportional

	SONAR		DIVE	
	0.06	0.05	0.06	0.07
	0.05	0.03	0.05	0.06
	0.03	0.008	0.03	0.05
	0.008	0e+00	0.008	0.03
	0.03	0.008	0.03	0.05

Inference in Homework 6: update

To compute the **posterior distribution**

$$P(T|C) = \frac{P(T) P(C|T)}{P(C)} \propto P(T) P(C|T)$$

1. Compute $P(T) \cdot P(C|T)$ for all possible treasure locations. To iterate over all possible locations, use the dictionary `self.current_distribution`.
2. Hint: store the products in `self.current_distribution`
3. Normalize (we have a single treasure and it must be in one location)
 - Add the products computed in 1 for all (x, y)
 - Hint: use *sum* and `self.current_distribution.values()`
 - Divide each value by the sum computed above

	SONAR	DIVE			
	0.06	0.05	0.06	0.07	0.08
	0.05	0.03	0.05	0.06	0.07
	0.03	0.008	0.03	0.05	0.06
	0.008	0e+00	0.008	0.03	0.05
	0.03	0.008	0.03	0.05	0.06

Inference in Homework 6: update

Prior distribution over treasure location: $P(T)$

- no longer uniform
- ▶ C = sonar color measured at $(4, 0)$
- ▶ $P(C = \text{yellow} \mid T)$ from model
- ▶ We can calculate the posterior distribution $P(T|C)$ over treasure locations given a sonar reading using Bayes' rule:

$$P(T|C) = \frac{P(T) P(C|T)}{P(C)} \propto P(T) P(C|T)$$

	SONAR		DIVE	
	0.05	0.05	0.09	0.1
	0.03	0.02	0.05	0.09
	0.007	0.004	0.02	0.05
	0e+00	0e+00	0.004	0.02
	0e+00	0e+00	0.007	0.03

Inference in Homework 6: update

To compute the **posterior distribution**

$$P(T|C) = \frac{P(T) P(C|T)}{P(C)} \propto P(T) P(C|T)$$

1. Compute $P(T) \cdot P(C|T)$ for all possible treasure locations. To iterate over all possible locations, use the dictionary `self.current_distribution`.
2. Hint: store the products in `self.current_distribution`
3. Normalize (we have a single treasure and it must be in one location)
 - Add the products computed in 1 for all (x, y)
 - **Hint: use *sum* and `self.current_distribution.values()`**
 - Divide each value by the sum computed above

	SONAR		DIVE	
	0.05	0.05	0.09	0.1
	0.03	0.02	0.05	0.09
	0.007	0.004	0.02	0.05
	0e+00	0e+00	0.004	0.02
	0e+00	0e+00	0.007	0.03

Homework 6: recommend_sensing

Return the position where we should take the next sonar reading in the grid.

1. the most promising (highest probability) unobserved location: *use `max` on the set of unobserved locations (`self.open`) with a `lambda` key*
2. If all remaining unobserved locations have a probability of 0, return the unobserved location that is closest to the location with the highest probability: *use the `closest_point` function in `utils`.*
3. If there are no remaining unobserved locations, return the location with the highest probability: *use `max` on the dictionary `self.current_distribution` with a `lambda` key.*

Time and Uncertainty

- ▶ The world changes, we need to track it and predict it
- ▶ Often, we want to **reason about a sequence** of observations
 - Robot localization
 - Medical monitoring
 - Speech recognition...
- ▶ Need to introduce time into our models

Basic Idea

State random variables for each time step

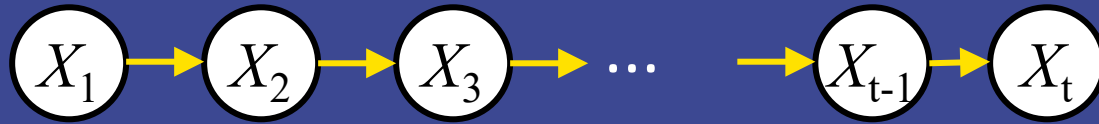
- X_t = set of state random variables at time t

State variables may be observable or unobservable

This assumes discrete time.

Bayes Net?

- Construct a Bayes net from these state variables?



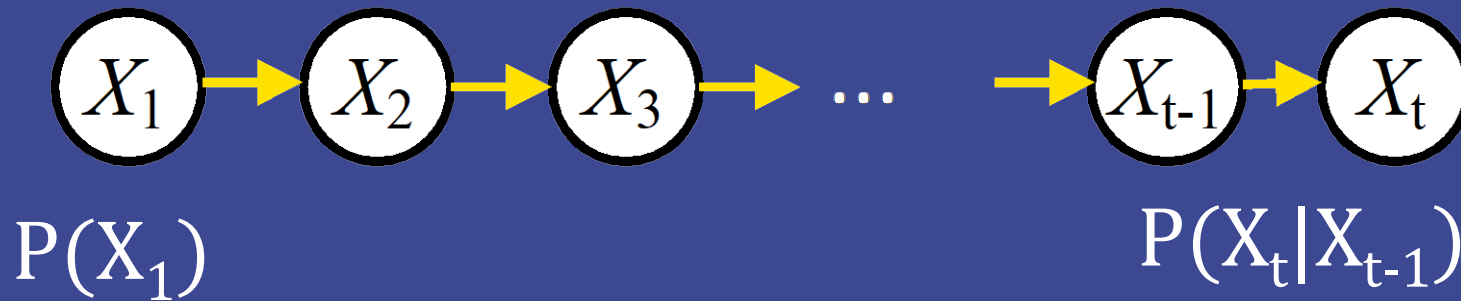
- Arcs / parents?
- Markov assumption: X_t depends only on a subset of $X_1, X_2, \dots, X_{t-1}$
 - First-order Markov: state X_t at each time step only depends on the previous state X_{t-1}

$$X_t \perp\!\!\!\perp X_1, \dots, X_{t-2} \mid X_{t-1}$$

$$P(X_t \mid X_{1:t-1}) = P(X_t \mid X_{t-1})$$

$$\text{Notation: } X_{1:t-1} = X_1, X_2, \dots, X_{t-1}$$

Markov Chain



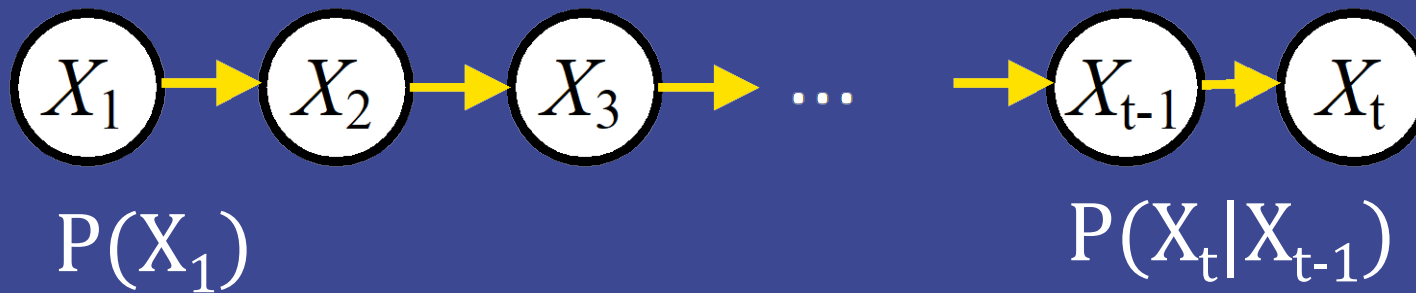
Parameters:

- initial state probabilities
- **transition probabilities** or dynamics, specify how the state evolves over time

Stationarity assumption: **transition probabilities are the same at all times**

$P(X_t | X_{t-1})$ is the same for all t

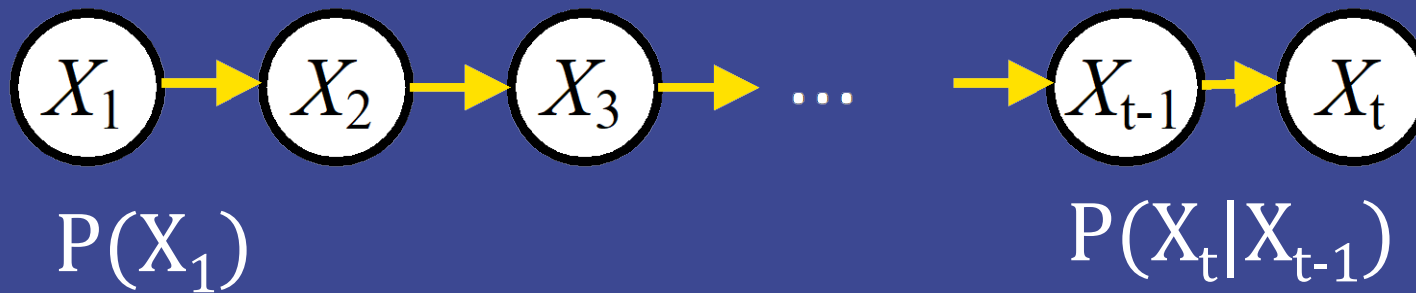
A Markov Chain is a Bayes Net



$$P(X_1, X_2, X_3) = ?$$

- A. $P(X_1) P(X_2) P(X_3)$
- B. $P(X_1|X_2)P(X_2|X_3) P(X_3)$
- C. $P(X_1)P(X_2|X_1) P(X_3|X_2)$
- D. $P(X_1) + P(X_2) + P(X_3)$

A Markov Chain is a Bayes Net



$$P(X_1, X_2, X_3) = P(X_1)P(X_2 | X_1) P(X_3 | X_2)$$

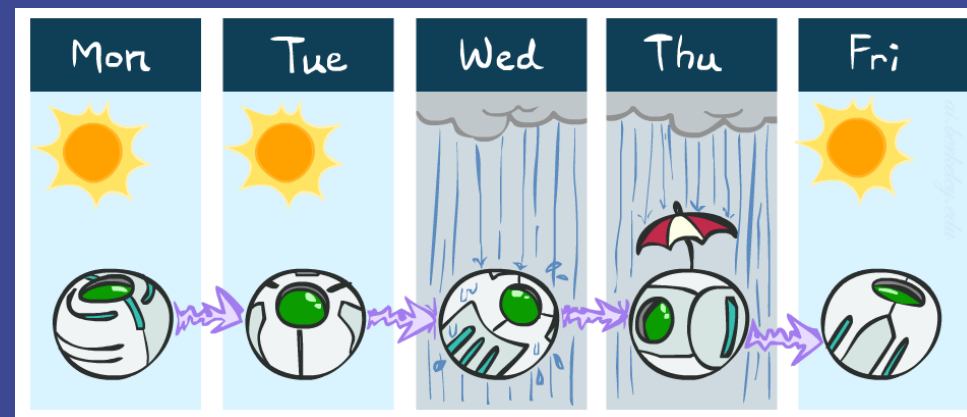
More generally:

$$P(X_1, X_2, \dots, X_t) = P(X_1)P(X_2 | X_1) \dots P(X_t | X_{t-1})$$

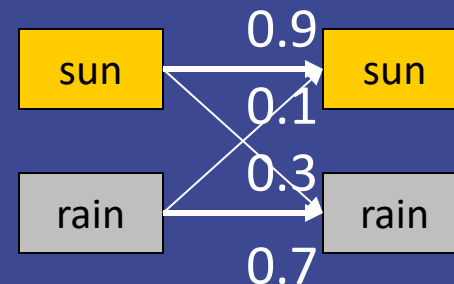
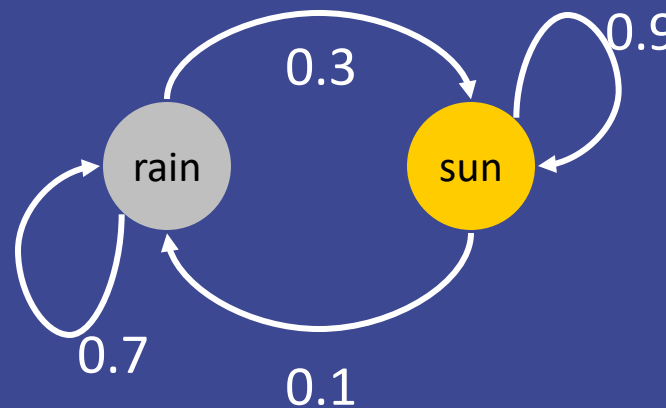
Example Markov Chain: Weather

- States: $X = \{\text{rain}, \text{sun}\}$
- Initial distribution: 1.0 sun
- CPT $P(X_t \mid X_{t-1})$:

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

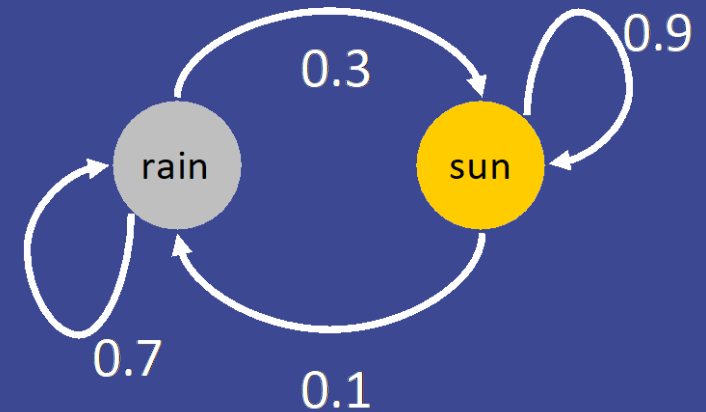


Two new ways of representing the same CPT



Example Markov Chain: Weather

- ▶ Initial distribution: $P(X_1 = \text{sun}) = 1.0$
- ▶ What is the probability distribution after one step/time unit?

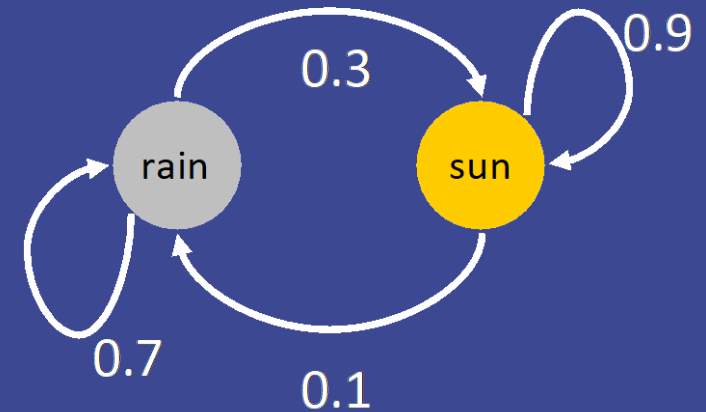


$$\begin{aligned} P(X_2 = \text{sun}) &= P(X_2 = \text{sun}, X_1 = \text{sun}) + P(X_2 = \text{sun}, X_1 = \text{rain}) \\ &= P(X_2 = \text{sun} \mid X_1 = \text{sun}) P(X_1 = \text{sun}) + \\ &\quad P(X_2 = \text{sun} \mid X_1 = \text{rain}) P(X_1 = \text{rain}) \\ &= 0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9 \end{aligned}$$

iClicker:
What is $P(X_2 = \text{rain})$?

Example Markov Chain: Weather

- ▶ Initial distribution: $P(X_1 = \text{sun}) = 1.0$
- ▶ What is the probability distribution after one step/time unit?

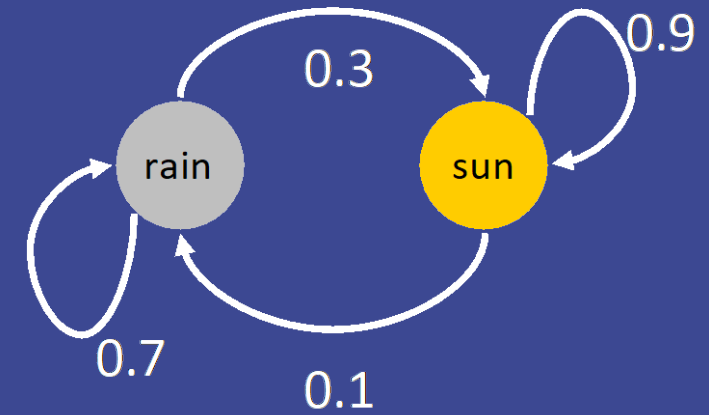


$$\begin{aligned} P(X_2 = \text{sun}) &= P(X_2 = \text{sun}, X_1 = \text{sun}) + P(X_2 = \text{sun}, X_1 = \text{rain}) \\ &= P(X_2 = \text{sun} \mid X_1 = \text{sun}) P(X_1 = \text{sun}) + \\ &\quad P(X_2 = \text{sun} \mid X_1 = \text{rain}) P(X_1 = \text{rain}) \\ &= 0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9 \end{aligned}$$

$$P(X_2 = \text{rain}) = 1 - 0.9 = 0.1$$

Example Markov Chain: Weather

- ▶ Initial distribution: $P(X_1 = \text{sun}) = 1.0$
- ▶ $P(X_2 = \text{sun}) = 0.9$
- ▶ $P(X_2 = \text{rain}) = 0.1$

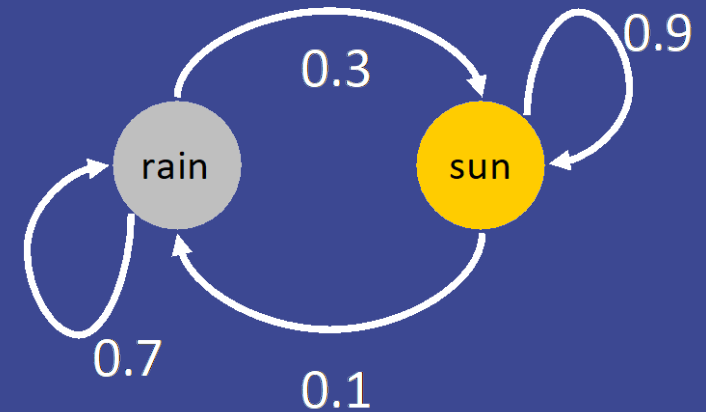


iClicker:

What is $P(X_3 = \text{sun})$?

Example Markov Chain: Weather

- ▶ Initial distribution: $P(X_1 = \text{sun}) = 1.0$
- ▶ $P(X_2 = \text{sun}) = 0.9$
- ▶ $P(X_2 = \text{rain}) = 0.1$
- ▶ What $P(X_3 = \text{sun})$?

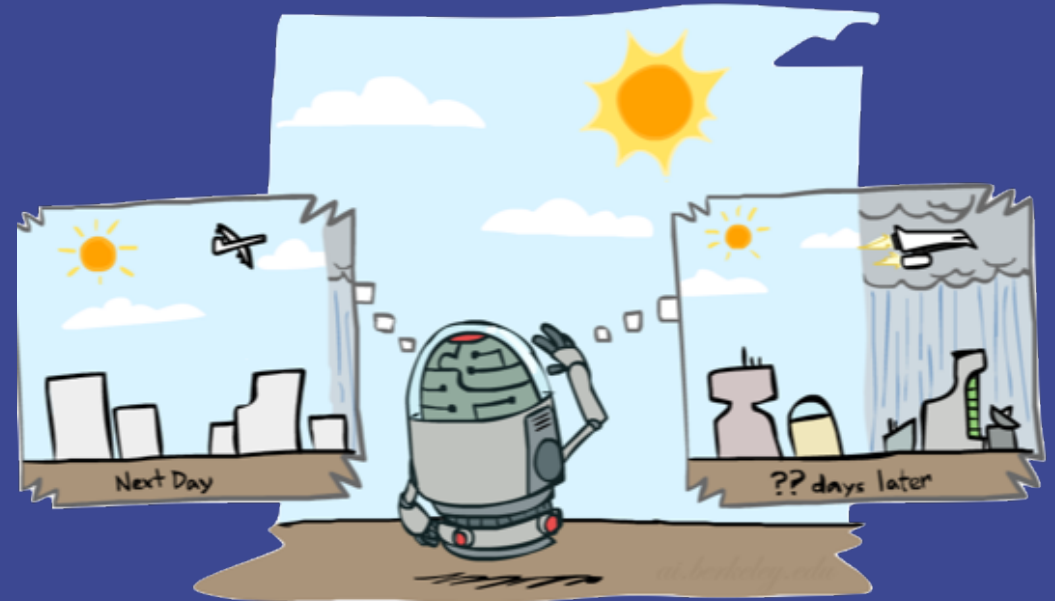
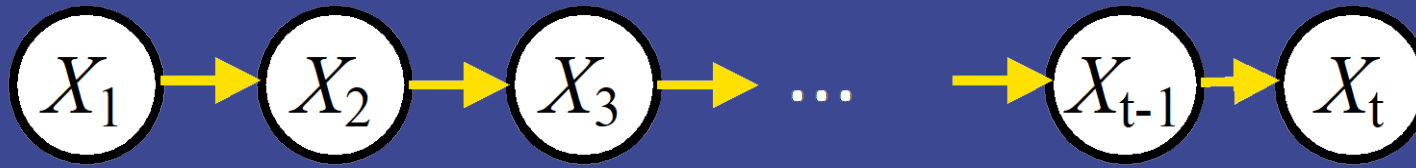


$$\begin{aligned} P(X_3 = \text{sun}) &= P(X_3 = \text{sun}, X_2 = \text{sun}) + P(X_3 = \text{sun}, X_2 = \text{rain}) \\ &= P(X_3 = \text{sun} \mid X_2 = \text{sun}) P(X_2 = \text{sun}) + \\ &\quad + P(X_3 = \text{sun} \mid X_2 = \text{rain}) P(X_2 = \text{rain}) \\ &= 0.9 \cdot 0.9 + 0.3 \cdot 0.1 = 0.84 \end{aligned}$$

$$P(X_3 = \text{rain}) = 1 - 0.84 = 0.16$$

Mini-Forward Algorithm

- ▶ Question: What's $P(X)$ on some day t ?



Example Run of Mini-Forward Algorithm

- From initial observation of sun

$$\begin{array}{cccc} \begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix} & \begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix} & \begin{bmatrix} 0.84 \\ 0.16 \end{bmatrix} & \begin{bmatrix} 0.804 \\ 0.196 \end{bmatrix} & \xrightarrow{\hspace{1cm}} & \begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix} \\ P(X_1) & P(X_2) & P(X_3) & P(X_4) & & P(X_\infty) \end{array}$$

- From initial observation of rain

$$\begin{array}{cccc} \begin{bmatrix} 0.0 \\ 1.0 \end{bmatrix} & \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix} & \begin{bmatrix} 0.48 \\ 0.52 \end{bmatrix} & \begin{bmatrix} 0.588 \\ 0.412 \end{bmatrix} & \xrightarrow{\hspace{1cm}} & \begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix} \\ P(X_1) & P(X_2) & P(X_3) & P(X_4) & & P(X_\infty) \end{array}$$

- From yet another initial distribution $P(X_1)$:

$$\begin{array}{ccc} \begin{bmatrix} p \\ 1 - p \end{bmatrix} & \xrightarrow{\hspace{1cm}} & \begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix} \\ P(X_1) & & P(X_\infty) \end{array}$$

Stationary Distributions

- ▶ For most chains:
 - Influence of the initial distribution diminishes over time.
 - The distribution we end up in is independent of the initial distribution
- ▶ Stationary distribution:
 - The distribution we end up with is called the **stationary distribution** of the chain
 - It satisfies: $P_{\infty}(X) = P_{\infty+1}(X)$



Example Markov Chain: Weather

$P(\text{sun})$ on November 4, 2045?

$$P_{\infty}(\text{sun}) = P_{\infty+1}(\text{sun})$$

$$\begin{aligned} P_{\infty}(\text{sun}) &= P(X_{\infty+1} = \text{sun}, X_{\infty} = \text{sun}) + P(X_{\infty+1} = \text{sun}, X_{\infty} = \text{rain}) \\ &= P(\text{sun} | \text{sun}) P_{\infty}(\text{sun}) + P(\text{sun} | \text{rain}) P_{\infty}(\text{rain}) \end{aligned}$$

$$P_{\infty}(\text{sun}) = 0.9 P_{\infty}(\text{sun}) + 0.3 P_{\infty}(\text{rain})$$

$$0.1 P_{\infty}(\text{sun}) = 0.3 P_{\infty}(\text{rain})$$

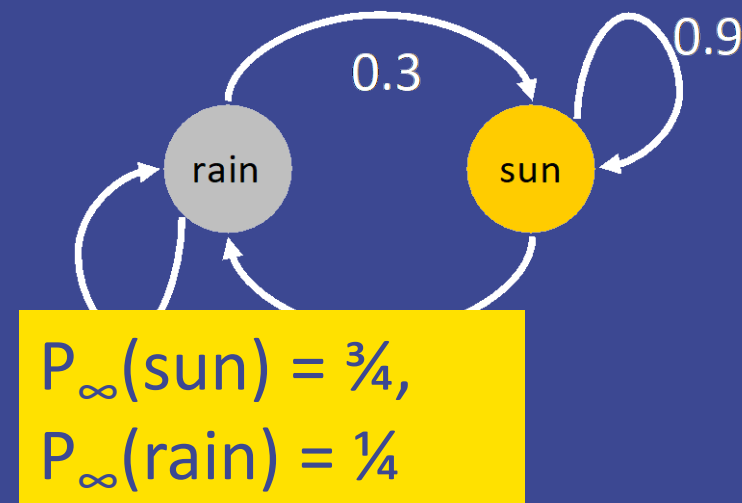
$$P_{\infty}(\text{sun}) = 3 P_{\infty}(\text{rain})$$

$$\begin{aligned} x &= 3y \\ x + y &= 1 \end{aligned}$$

We also know that: $P_{\infty}(\text{sun}) + P_{\infty}(\text{rain}) = 1$

$$y = 1 - x$$

$$x = 3(1 - x) \Rightarrow x = 3 - 3x \Rightarrow x = \frac{3}{4}, y = \frac{1}{4}$$



Hidden Markov Models

For most agents, we have both:

- ▶ **transition probabilities** or dynamics: specify how the state evolves over time
- ▶ **observations/evidence** to update our beliefs

Basic Idea

State and **evidence** random variables **for each time step**

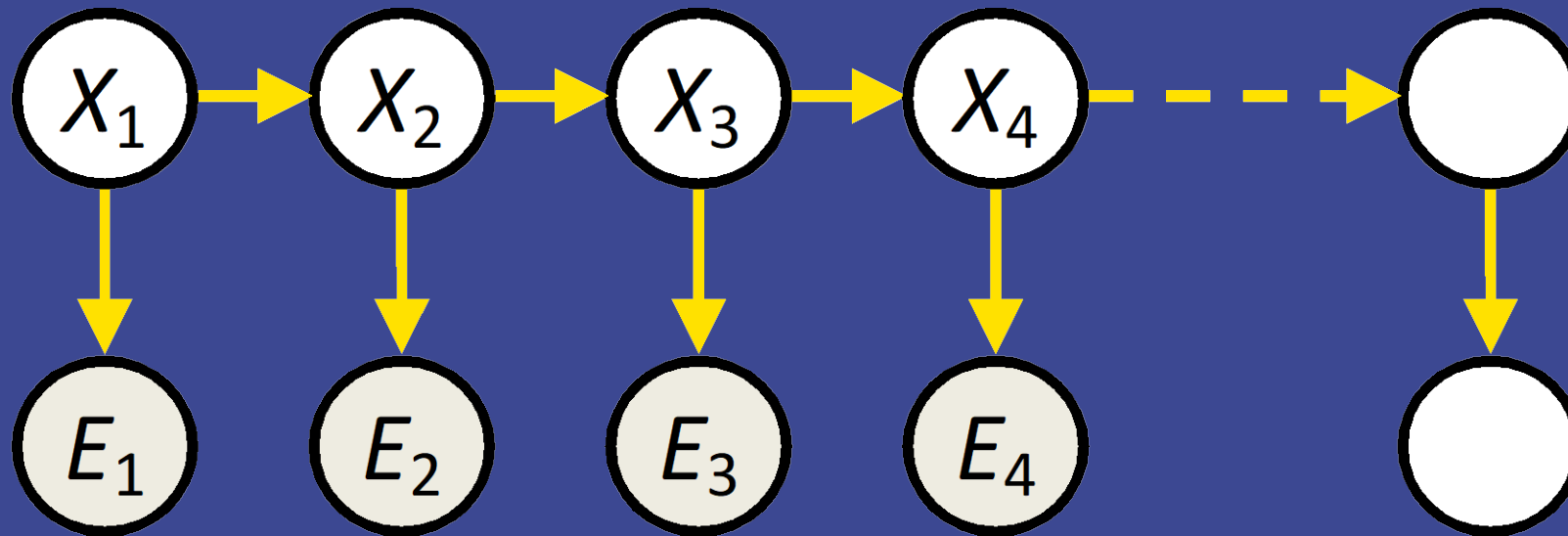
- X_t = set of state random variables at time t

State variables are **unobservable**

- E_t = set of observable evidence variables at time t

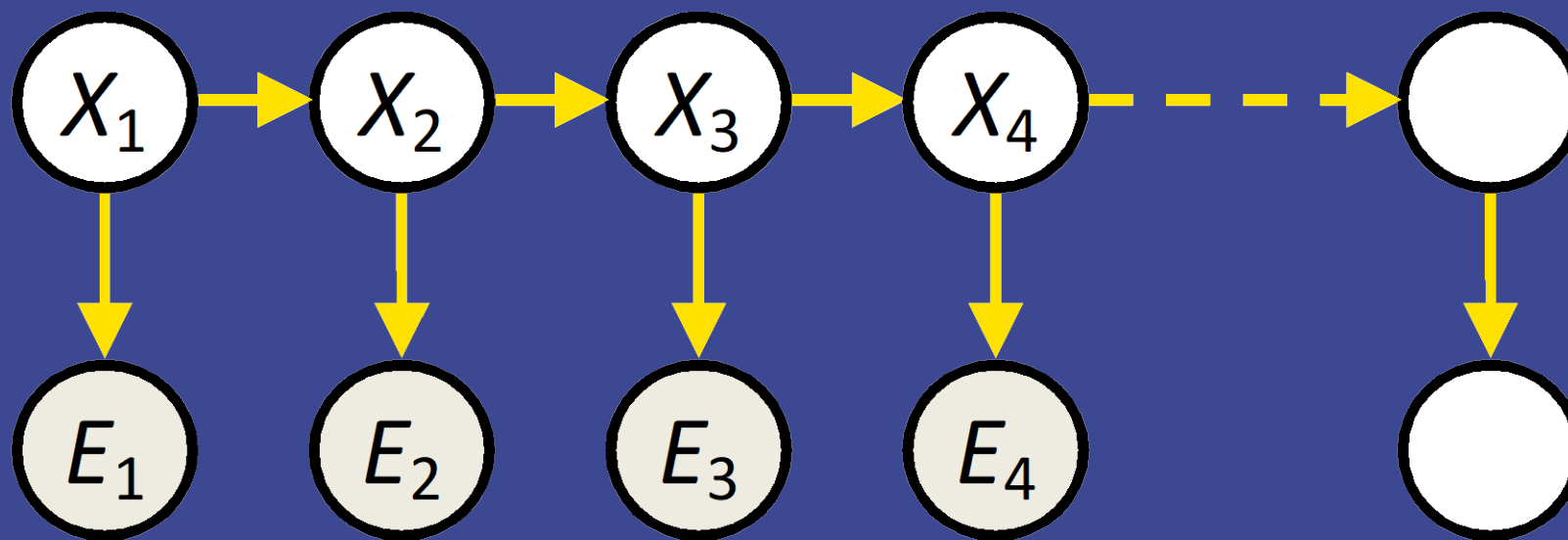
Hidden Markov Models

- Underlying Markov chain over states X : transition model: $P(X_t | X_{t-1})$
- We observe evidence at each time step: emissions/sensor model: $P(E_t | X_t)$

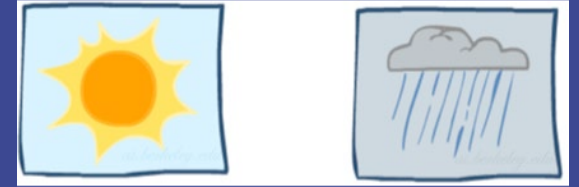
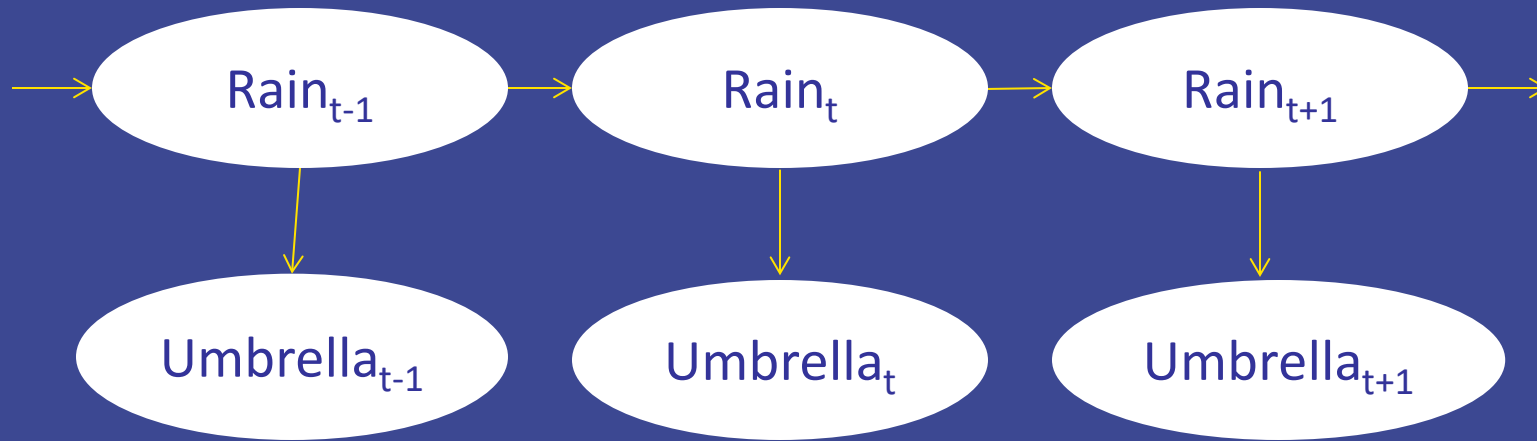


Hidden Markov Models

- ▶ An HMM is defined by:
 - Initial distribution: $P(X_1)$
 - Transitions: $P(X_t | X_{t-1})$
 - Emissions: $P(E_t | X_t)$

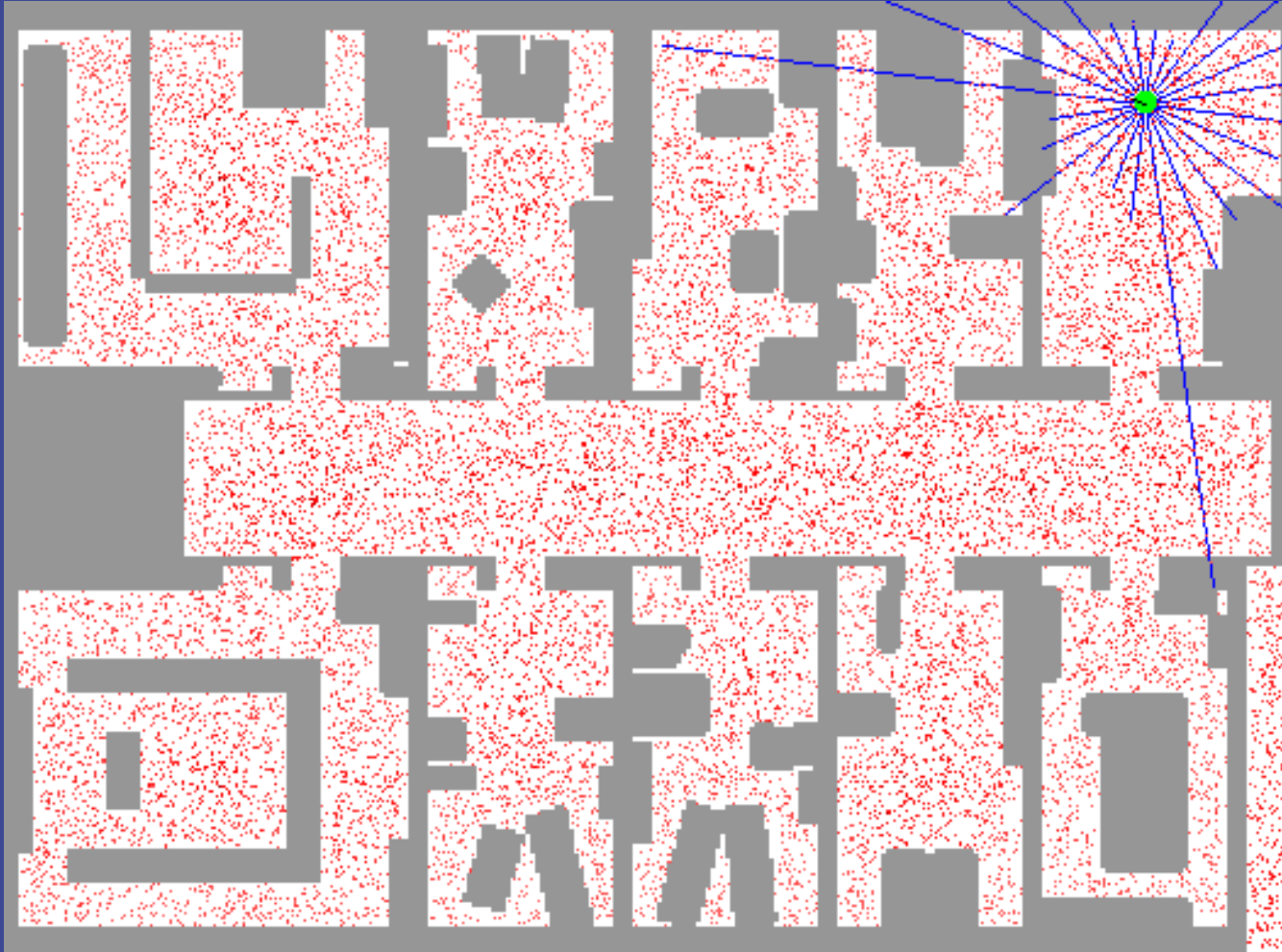


Example: Weather HMM

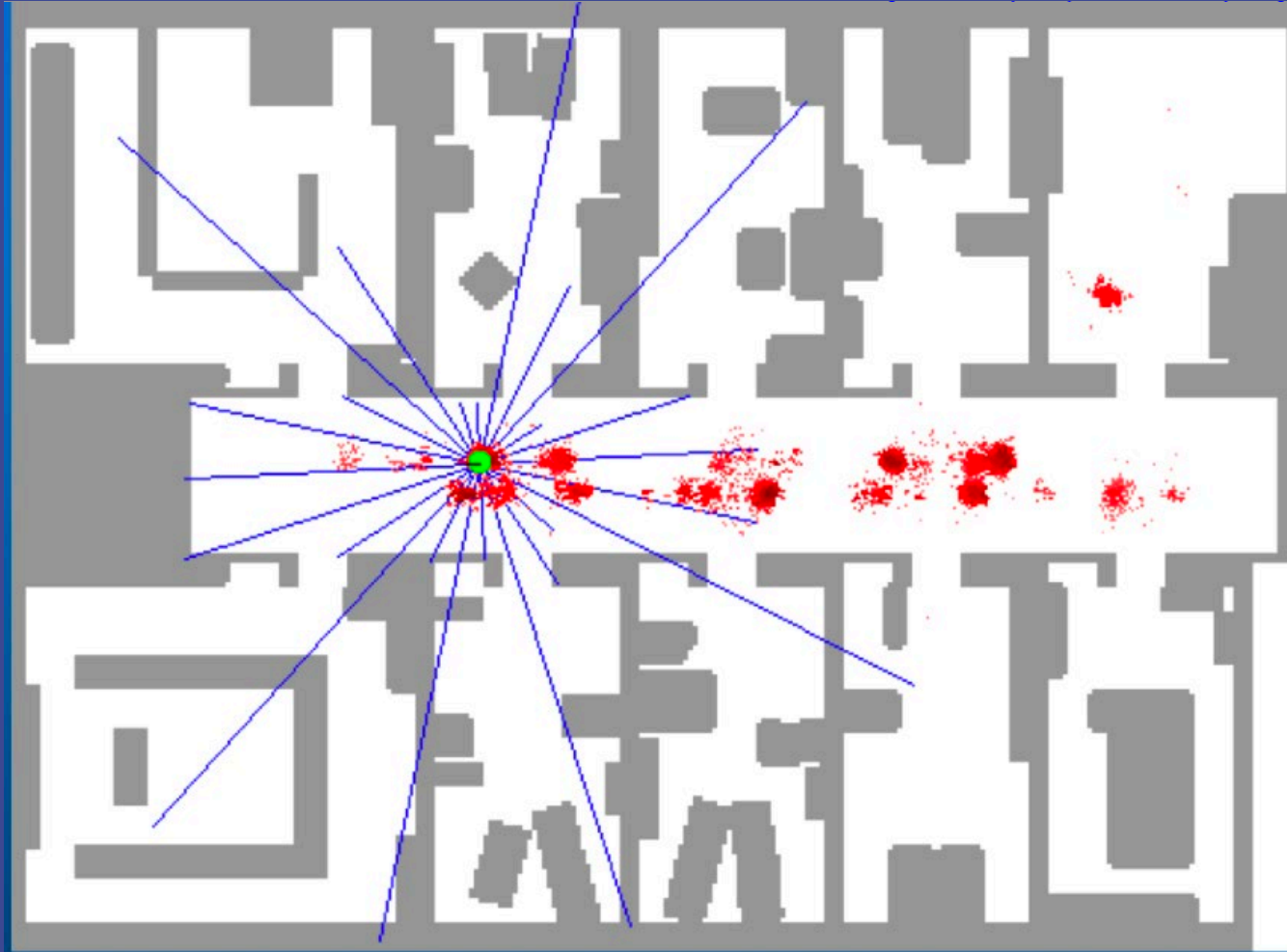


- ▶ An HMM is defined by:
 - Initial distribution: $P(X_1)$
 - Transitions: $P(X_t | X_{t-1})$
 - Emissions: $P(U_t | X_t)$

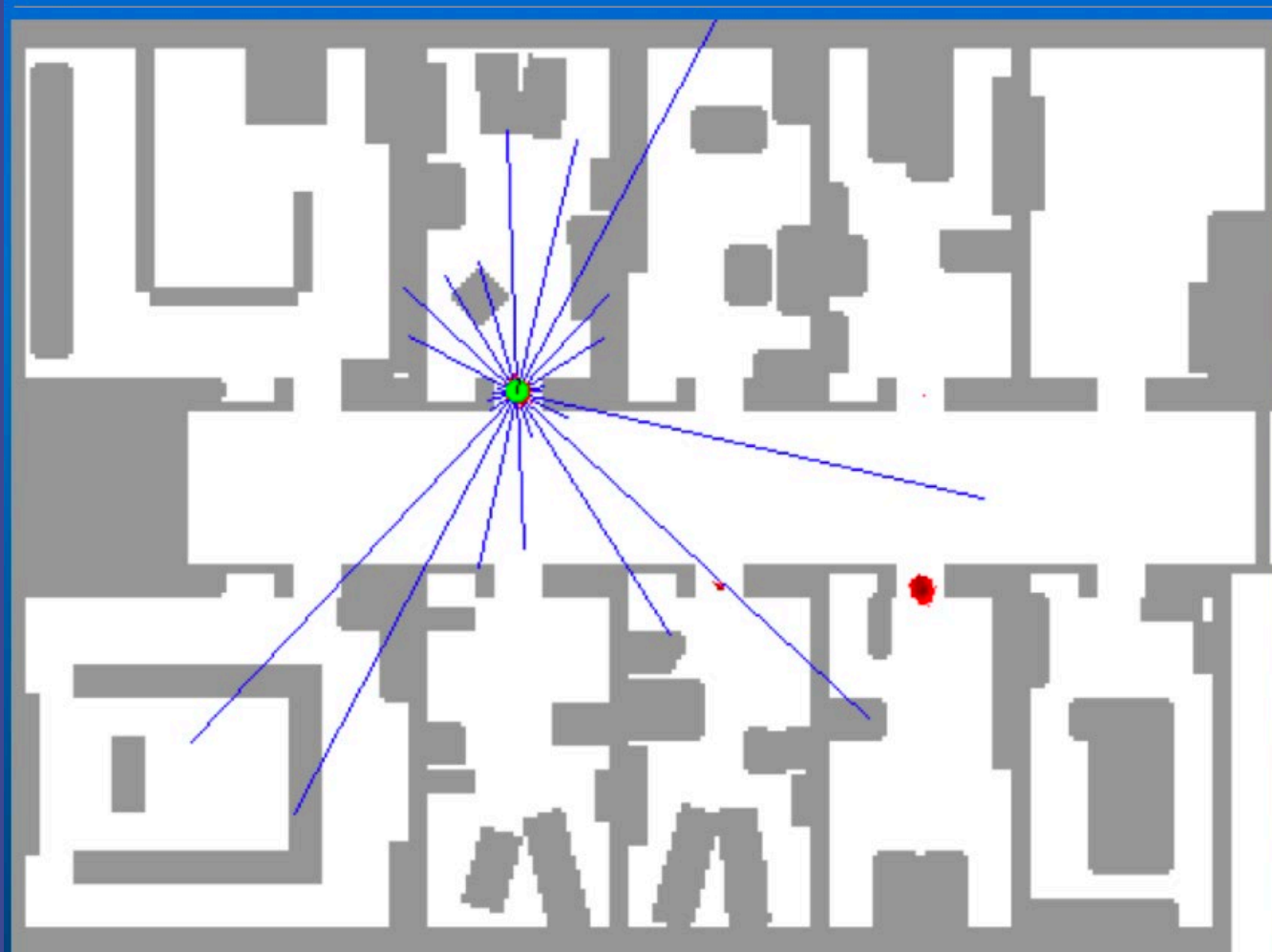
Example: Robot Localization



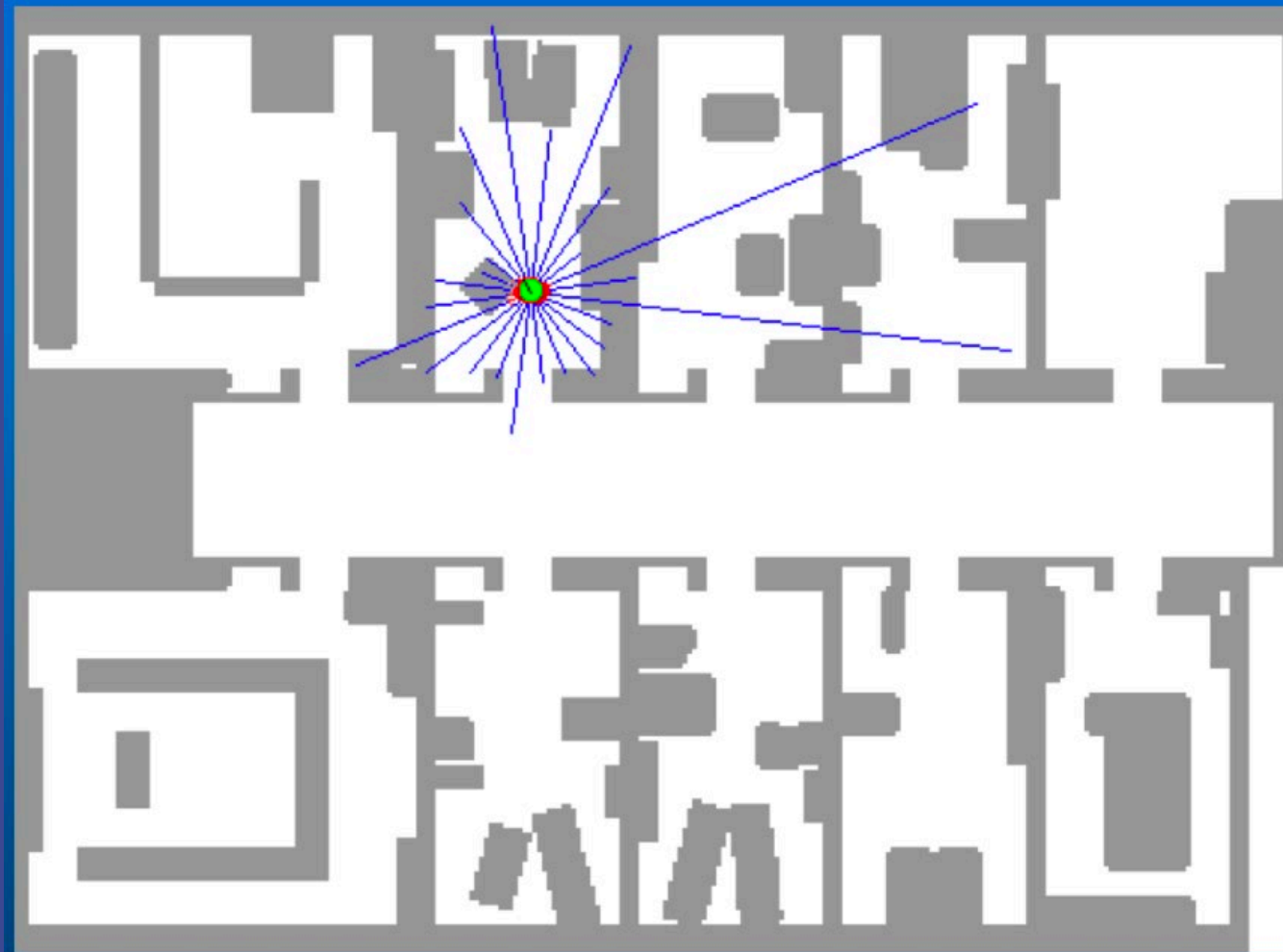
Example: Robot Localization



Example: Robot Localization



Example: Robot Localization



Reminders

- ▶ Homework 6: due November 13
- ▶ Questions? Please ask!